



Fairness Verification Checklist

Quick bias detection toolkit



5 Bias Detection Techniques

Use these techniques to proactively identify potential biases in your AI systems. Remember, fairness is context-dependent, and the best approach combines multiple methods.

1. Diverse Test

Test your AI across diverse demographic groups. Ensure the training data represents the population it will serve.

Checklist:

- ☐ Data includes diverse representation.
- ☐ Testing covers various demographic segments.
- ☐ Performance metrics are disaggregated by group.

Example Questions:

- Does the dataset reflect the real-world distribution of the target population?
- Are there any protected characteristics that are underrepresented or overrepresented?
- How does the model perform across different demographic subgroups?

2. Flip Test

Reverse key inputs to check for unintended bias amplification. For example, swap genders, names, or ethnicities.

Checklist:

- ☐ Key inputs have been systematically reversed.
- ☐ Output changes are carefully analyzed.
- ☐ Unintended bias amplification is addressed.

Example Questions:

- What happens when you change the gender associated with a resume?
- Does the model exhibit different behavior for names associated with different ethnic groups?
- Are there any unexpected or discriminatory outcomes when inputs are 'flipped'?

3. Explicit Anti-Bias Prompting

Use prompts specifically designed to mitigate bias. For example, ask for explanations or justifications for decisions.

Checklist:

- ☐ Anti-bias prompts are integrated into the system.
- ☐ Prompt responses are analyzed for bias.
- ☐ Prompts are iteratively refined.

Example Questions:

- Can the model explain its reasoning behind a particular decision?
- Does the model's explanation reveal any biases or stereotypes?
- Are there any patterns in the explanations that suggest unfair treatment?

4. Multiple Perspectives

Engage stakeholders with diverse backgrounds and experiences to review and evaluate the AI system.

Checklist:

- ☐ Diverse stakeholders are involved in the review process.
- ☐ Stakeholder feedback is actively solicited and incorporated.
- ☐ Different perspectives are considered during evaluation.

Example Questions:

- Do different stakeholders perceive the system as fair?
- Are there any potential biases that are not immediately apparent?
- What are the potential unintended consequences of the system?

5. Diverse Review Team

Assemble a team with varied expertise and backgrounds to comprehensively assess the AI's fairness.

Checklist:

- ☐ The review team includes diverse representation.

- ☐ Team members have expertise in relevant areas.
- ☐ Review findings are documented and addressed.

Example Questions:

- Does the review team have the necessary skills and knowledge to assess fairness?
- Are there any conflicts of interest among team members?
- How will the review findings be used to improve the system?

Accountability Framework

Establish a clear accountability framework to ensure fairness is prioritized and maintained throughout the AI lifecycle.

- ☐ **Verification:** Regularly verify the AI system's fairness using the techniques above. *(Ensures ongoing monitoring and detection of biases.)*
- ☐ **Documentation:** Maintain comprehensive documentation of the AI system's development, testing, and deployment. *(Provides transparency and facilitates auditing.)*
- ☐ **Override Authority:** Establish a clear process for overriding the AI system's decisions when necessary. *(Allows for human intervention in cases of unfair outcomes.)*
- ☐ **Transparency:** Be transparent about the AI system's capabilities and limitations. *(Builds trust and enables informed decision-making.)*

Bias Red Flags

Watch out for these red flags that may indicate potential bias in your AI system:

- ☐ **Disparate Impact:** Different outcomes for different groups.
- ☐ **Data Skew:** Training data doesn't reflect reality.
- ☐ **Feedback Loops:** Biased outputs reinforce biased inputs.
- ☐ **Lack of Transparency:** Difficulty understanding how decisions are made.
- ☐ **Over-reliance:** Automating decisions without human oversight.

Quick Assessment Table

AI Use Case	Tests Applied	Bias Found?	Actions Taken

Note: Use this table to document your fairness verification process for each AI use case.

Final Thoughts

Fairness is not a technological problem alone; it's a **human responsibility**. Use this toolkit to guide your efforts, but always remember to apply critical thinking and ethical judgment.

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